Milestone 3: Data Pipeline & Initial Model

Weekly Sales Forecasting Walmart Dataset (2010–2012)

AIE1014 - AI Applied Project Course

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1. Executive Summary

This project develops a weekly retail sales forecasting framework using the Walmart dataset (2010–2012), combining statistical and machine learning methods to predict total weekly store sales.

The dataset contains 6,435 rows and 8 columns covering 45 stores over 143 weeks (Feb 2010 – Oct 2012). The target variable is Weekly_Sales (continuous). Key features include Store, Temperature, Fuel_Price, CPI, Unemployment, and Holiday_Flag.

The Random Forest model achieved a test RMSE of 1,648,722 and MAE of 951,806, representing improvements of 33.5% and 48.8% respectively over the naive baseline (RMSE: 2,481,007 / MAE: 1,859,968), comfortably exceeding the 10% threshold required for this milestone.

2. Data Pipeline

2.1 Data Collection

Source: Walmart Store Sales dataset (Kaggle), loaded from a local CSV file using pandas. The dataset contains 6,435 rows and 8 columns (approximative0.75 MB), requiring no memory optimization. Columns: Store (int), Date (object), Weekly_Sales (float, target), Holiday_Flag (int), Temperature, Fuel_Price, CPI, Unemployment (all float).

2.2 Data Exploration

Weekly_Sales has a mean of approximative1.05M (std approximative564K), is right-skewed (IQR: 553K–1.42M), with extreme values up to 3.82M. IQR-based outlier detection identified 34 values beyond thresholds. After weekly aggregation (W-FRI), the series spans 143 weeks with no missing values. The ADF test confirmed stationarity (statistic: -5.91, p=0.00). The Ljung-Box test confirmed significant autocorrelation (p=0.000008). The ACF plot showed a strong seasonal spike near lag 52, confirming annual periodicity.

2.3 Data Preprocessing

No missing values were found across all columns. The 34 IQR outliers were retained as they represent genuine holiday-driven sales spikes. A causal Hampel filter (window=7, threshold=3σ) was applied to the weekly series to replace remaining outliers using past data only. Mixed date formats (DD/MM/YYYY and MM/DD/YYYY) were resolved via a two-pass pd.to_datetime strategy. The Store column was encoded using OneHotEncoder (handle_unknown='ignore', sparse_output=False).

2.4 Feature Engineering

Lag features: y_lag_1, 2, 4, 8, 12, 26, 52. Rolling features (causal, shift(1)): ma_4, ma_12, std_4. Calendar features: weekofyear, month, year. Mutual information ranking on the training set identified the top predictors: y_lag_52 (2.004), ma_4 (1.875), Store (1.829), ma_12 (1.767), y_lag_1 (1.742). Calendar features scored 0.000 (redundant, captured by lags). Final feature set: 14 features total.

2.5 Train/Test Split

Chronological split with cutoff January 1, 2012: 48 training samples / 43 test samples (approximative53%/47%). No stratification (time series regression). StandardScaler fitted on training data only and applied to test set. Processed splits saved to CSV files and reused consistently across notebooks to ensure no leakage.

3. Baseline Model

The Naive (Last Value) baseline predicts the last observed training value for all test dates: ŷt = ytrain[−1] ∀t ∈ test. This represents the simplest possible forecasting strategy, capturing no trend or seasonality, and sets a meaningful lower bound for model performance.

Metric	Baseline Value
RMSE	2,481,006.64
MAE	1,859,967.56

Any model failing to outperform this baseline provides no practical forecasting value, making it an appropriate and rigorous benchmark for this regression task.

4. Initial Model

4.1 Model Selection

Two models were evaluated. SARIMA: m=52 was tested but is infeasible with only 48 training points (requires several hundred observations). m=12 was chosen as a pragmatic approximation. Auto_arima selected ARIMA(1,1,0)(1,0,0)_12, which yielded RMSE: 5,939,295 and MAE: 5,353,532 worse than the baseline with a near-singular covariance warning, confirming the dataset is better suited to machine learning. Random Forest was selected as the primary model for its ability to handle non-linear relationships, tabular lag features, and store-level heterogeneity.

4.2 Training Process

Features loaded from processed CSVs (same chronological split as pipeline). Log1p transformation applied to target before training; predictions converted back with expm1. Initial config: n_estimators=200, max_depth=10, random_state=42. GridSearchCV (3-fold CV) tuned: n_estimators∈[50,100,200], max_depth∈[None,10,20], min_samples_split∈[2,5,10]. Best config: max_depth=None, min_samples_split=10, n_estimators=200 (CV RMSE: 1,789,520). Feature reduction (top 80% by importance) also tested: RMSE 1,809,258.

4.3 Evaluation Results

Metric	Naive Baseline	SARIMA (m=12)	Random Forest (tuned)	Random forest Improvement
RMSE	2,481,006.64	5,939,294.72	1,648,722.36	−33.55%
MAE	1,859,967.56	5,353,531.52	951,806.03	−48.83%

4.4 Results Interpretation

The Random Forest significantly outperforms the naive baseline (RMSE −33.5%, MAE −48.8%), comfortably exceeding the 10% milestone threshold. Feature importance confirms lag_52 (approximative63%) as the dominant predictor, reflecting strong annual seasonality, followed by lag_2 (approximative24%) and lag_1 (approximative17%).

Overfitting analysis: train RMSE 1,278,301 vs. test RMSE 1,808,222 (gap ratio approximately 1.41x), indicating moderate overfitting attributable to the small training set (48 points). The close alignment between CV RMSE (1,789,520) and test RMSE (1,808,222) validates consistent generalization. The model captures genuine predictive structure rather than memorizing training data.

5. Challenges & Solutions

Challenge	Solution
Mixed date formats (DD/MM/YYYY vs MM/DD/YYYY)	Two-pass pd.to_datetime (dayfirst=True then False). All 143 dates resolved.
SARIMA m=52 infeasible on 48 training points	Used m=12 as pragmatic approximation; poor results confirmed ML superiority.
Risk of data leakage	StandardScaler fit on train only; causal rolling (shift(1)); chronological split enforced.
Inconsistent split between notebooks	Corrected by loading processed CSVs in model notebook identical cutoff guaranteed.
Moderate overfitting in Random Forest	GridSearchCV tuning (min_samples_split=10) + feature reduction improved generalization.

6. Next Steps — Planned Improvements for Milestone 4

6.1 Generate Synthetic Data

Time series augmentation techniques will be applied to increase training set size and diversity, enabling deep learning models to learn temporal patterns more effectively and address the current 48-point training limitation.

6.2 Additional Models

- LSTM and GRU (RNNs) for sequential dependencies in weekly sales.
- CNN for local temporal pattern extraction.
- Hybrid CNN-LSTM combining local pattern detection and sequential memory.
- Performance comparison against current Random Forest baseline.

6.3 Feature Engineering Ideas

- Additional lag features (for example lag_3, lag_6, lag_16) and wider rolling windows.
- Enhanced calendar effects: holiday indicators, promotional periods, seasonal dummies.
- External factors if available: weather data, marketing campaign indicators.

6.4 Timeline

Period	Tasks
Week 7–8	Generate synthetic data; preprocess for deep learning.
Week 8–9	Train and validate LSTM, GRU, CNN, and hybrid models.
Week 10	Evaluate, compare with baseline and Random Forest, interpret results.
Week 10	Finalize model selection, document results, submit Milestone 4.